

Chapter 6:

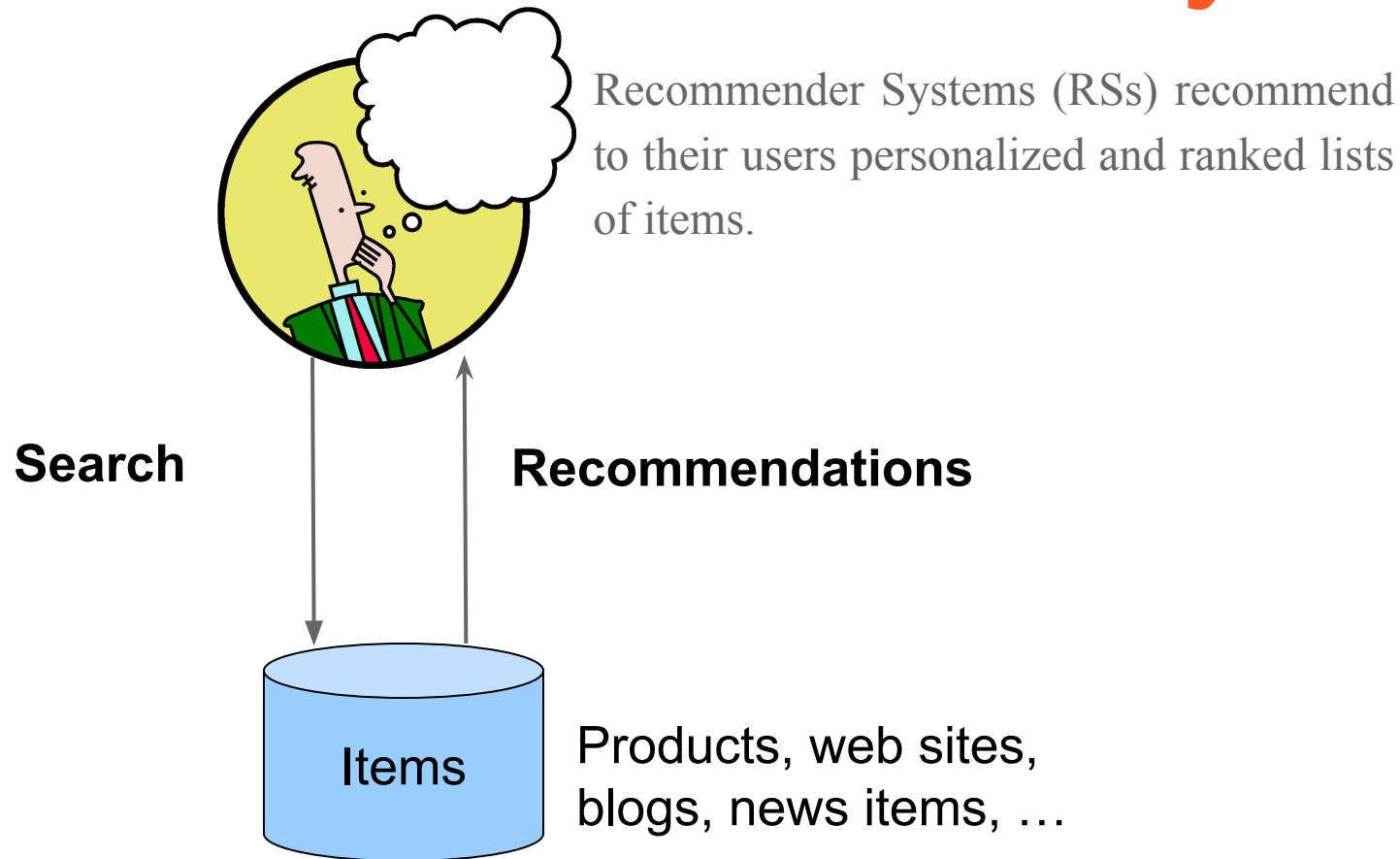
Recommendation Systems



Outlines

1. Introduction to recommender systems
2. A Model for Recommendation Systems
3. Content-based filtering
4. Collaborative filtering

Introduction to recommender systems



amazon.com.



Google
News

You Tube

Introduction to recommender systems

Two main groups:

- Content-based systems examine properties of the items recommended.
 - For instance, if a Netflix user has watched many cowboy movies, then recommend a movie classified in the database as having the “cowboy” genre.
- Collaborative filtering systems recommend items based on similarity measures between users and/or items. The items recommended to a user are those preferred by similar users.

A Model for Recommendation Systems

A Model for Recommendation Systems

The Utility Matrix

- Two classes of entities: *users* and *items*.
- *Users* have preferences for certain *items*, and these preferences must be teased out of the data.
- The data itself is represented as a *utility matrix*, giving for each user-item pair, a value represents the degree of preference of that user for that item.

A Model for Recommendation Systems

The Utility Matrix

The goal of a recommendation system is to predict the blanks in the utility matrix.

Example:

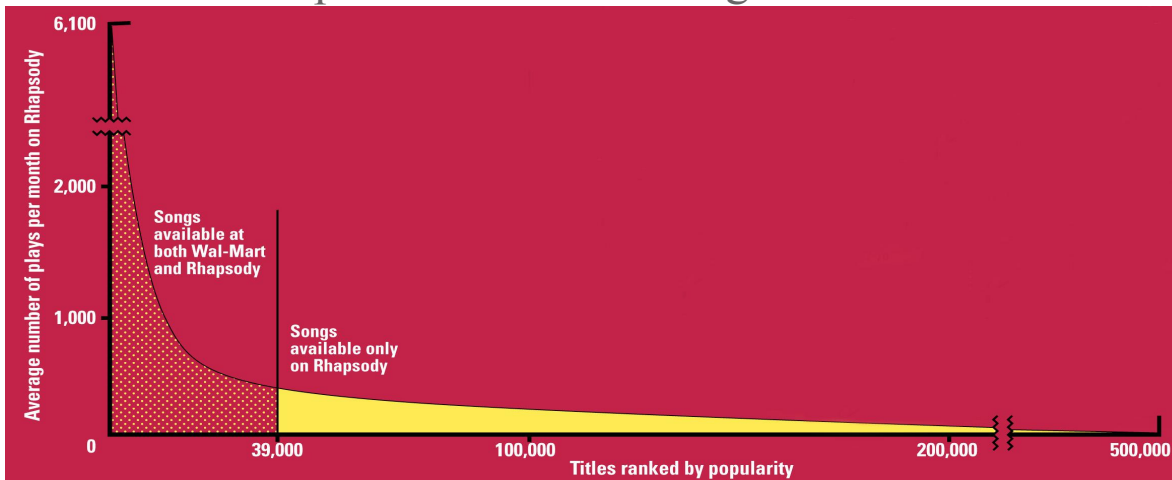
- A utility matrix represent users' ratings of movies on a 1–5 scale, with 5 the highest rating. Blanks represent the situation where the user has not rated the movie.
 - The movie names are HP1, HP2, and HP3 for Harry Potter I, II, and III, TW for Twilight, and SW1, SW2, and SW3 for Star Wars episodes 1, 2, and 3.
- The users are represented by capital letters A through D.

	HP1	HP2	HP3	TW	SW1	SW2	SW3
A	4			5	1		
B	5	5	4				
C				2	4	5	
D		3					3

A Model for Recommendation Systems

The Long Tail

- Recommendation in the physical world is fairly simple.
 - A bookstore will display only the books that are most popular
 - A newspaper will print only the articles it believes the most people will be interested in.
- Physical institutions provide only the most popular items to the left of the vertical line, while the corresponding online institutions provide the entire range of items: the tail as well as the popular items.



A Model for Recommendation Systems

Populating the Utility Matrix

Acquiring data from which to build a utility matrix is often difficult.

Two general approaches discover the value users place on items.

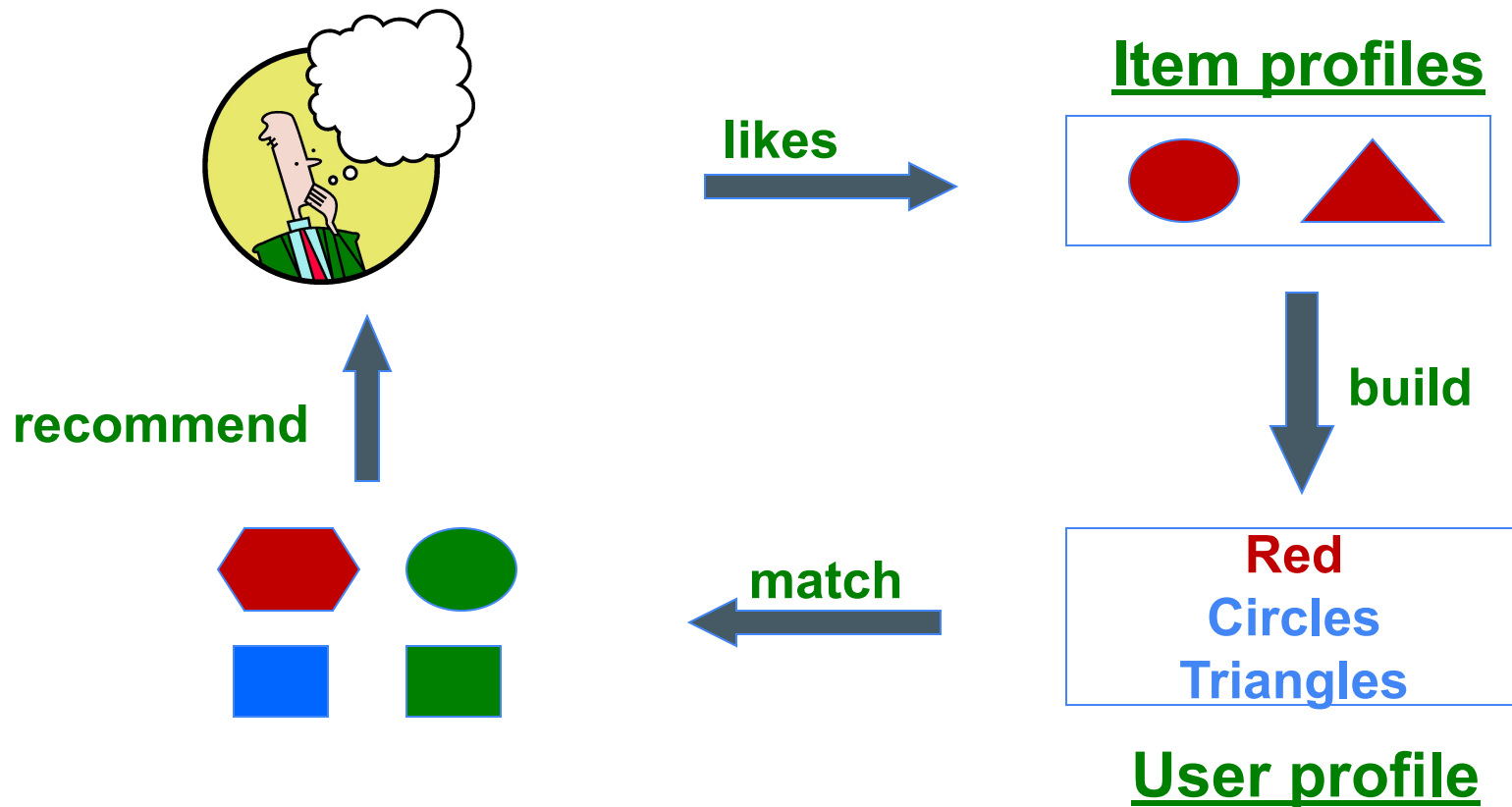
- We can ask users to rate items.
 - This approach is limited in its effectiveness, since generally users are unwilling to provide responses, and the information from those who do may be biased by the very fact that it comes from people willing to provide ratings.
- We can make inferences from users' behavior.
 - If a user buys a product at Amazon or views information about an item, watches a movie on YouTube, or reads a news article, then the user can be said to “like” this item.

Content-Based Recommendations

Content-Based Recommendations

- Content-Based systems focus on properties of items. Similarity of items is determined by measuring the similarity in their properties.
- Main idea:
 - Recommend items to customer x similar to previous items rated highly by x
- Example:
 - Movie recommendations
 - Recommend movies with same actor(s), director, genre, ...
 - Websites, blogs, news
 - Recommend other sites with “similar” content

Content-Based Recommendations



Content-Based Recommendations

Item Profiles: collections of records representing important characteristics of that item.

- Profile is a set (vector) of features
 - Movies: author, title, actor, director,...
 - Text: set of “important” words in document
- How to pick important features?
 - Usual heuristic from text mining is TF-IDF
 - (Term frequency * Inverse Doc Frequency)

Content-Based Recommendations

Discovering Features of Documents

- Eliminate stop words.
- Compute the TF.IDF score for each word in the document.
- Take the n words with the highest TF.IDF scores as features of the document.
- To measure the similarity of two documents, there are several natural distance measures:
 - Jaccard distance
 - cosine distance

Content-Based Recommendations

Obtaining Item Features From Tags

- Obtaining information about features of items by inviting users to tag the items by entering words or phrases that describe the item.
- The problem with tagging as an approach to feature discovery is that the process only works if users are willing to take the trouble to create the tags, and there are enough tags that occasional erroneous ones will not bias the system too much

Content-Based Recommendations

User profile:

- User has rated items with profiles $i_1, \dots, i_n$
- Simple: (weighted) average of rated item profiles
- Variation: normalize weights by difference from average rating of user
- Some other sophisticated aggregations

Content-Based Recommendations

User Profiles

	M1	M2	M3
U1	7	4	10

	Fantasy	Action	Cartoon	Drama	Comedy
M1	0	0	1	0	1
M2	0	0	0	1	0
M3	1	0	0	0	1

User profile

	Fantasy	Action	Cartoon	Drama	Comedy
U1					

Content-Based Recommendations

User Profiles

	U1
M1	7
M2	4
M3	10

	Fantasy	Action	Cartoon	Drama	Comedy
M1	0	0	1	0	1
M2	0	0	0	1	0
M3	1	0	0	0	1

	Fantasy	Action	Cartoon	Drama	Comedy
M1	0	0	7	0	7
M2	0	0	0	4	0
M3	10	0	0	0	10

	Fantasy	Action	Cartoon	Drama	Comedy
M1	10	0	7	4	17 ¹⁸

User profile

Content-Based Recommendations

Recommending Items to Users Based on Content

Prediction heuristic: Cosine similarity of user and item profiles

Given user profile x and item profile i , estimate

$$u(x, i) = \cos(x, i) = \frac{x \cdot i}{||x|| \cdot ||i||}$$

Content-Based Recommendations

Recommending Items to Users Based on Content

Prediction heuristic: Cosine similarity of user and item profiles

- Given user profile x and item profile i , estimate

	Fantasy	Action	Cartoon	Drama	Comedy
M4	1	0	0	0	1
M5	0	1	0	1	0

	Fantasy	Action	Cartoon	Drama	Comedy
M1	0.26	0	0.18	0.11	0.45

	Fantasy	Action	Cartoon	Drama	Comedy
M1	10	0	7	4	17

Content-Based Recommendations

Pros: Content-based Approach

- + No need for data on other users

 - No cold-start or sparsity problems

- + Able to recommend to users with unique tastes

- + Able to recommend new & unpopular items

 - No first-rater problem

- + Able to provide explanations

 - Can provide explanations of recommended items by listing content-features that caused an item to be recommended

Content-Based Recommendations

Cons: Content-based Approach

- : Finding the appropriate features is hard

E.g., images, movies, music

- : Recommendations for new users

How to build a user profile?

- : Overspecialization

Never recommends items outside user's content profile

People might have multiple interests

Unable to exploit quality judgments of other users

Collaborative filtering

Collaborative filtering

Find similar users and recommend items that they like.

Comes in two flavors:

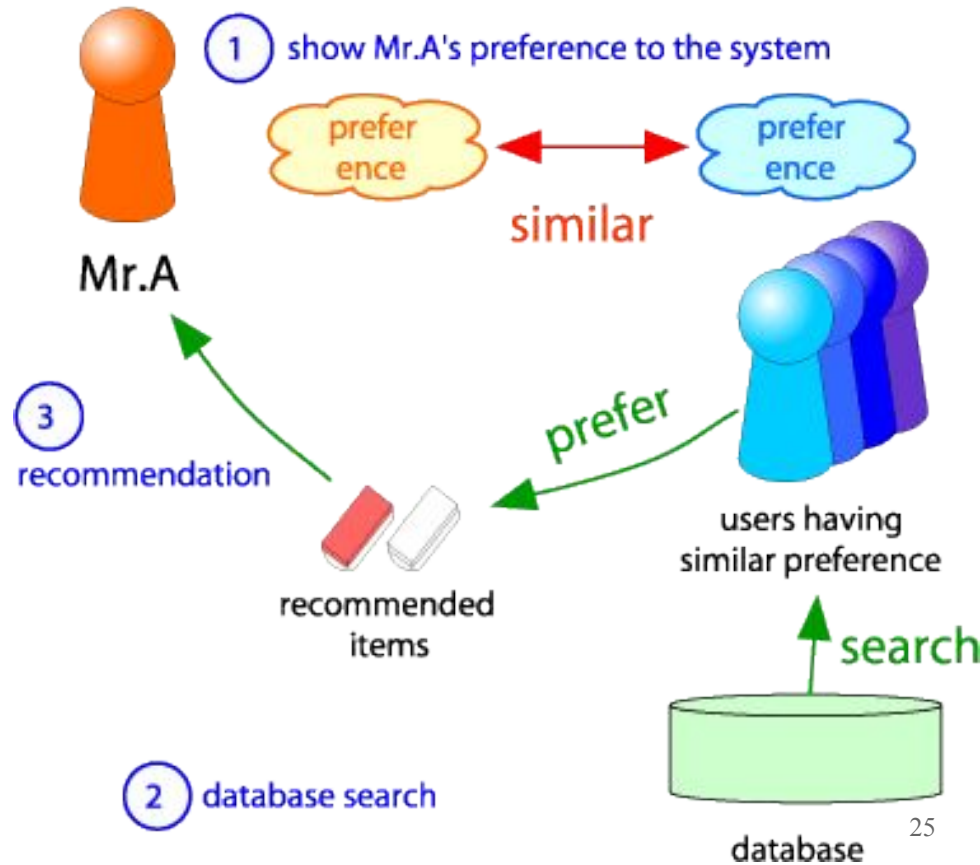
- User-user collaborative filtering

- Item-Item collaborative filtering

User-user collaborative filtering

Consider user x

- Find set N of other users whose ratings are “similar” to x’s ratings
- Estimate x’s ratings based on ratings of users in N



User-user collaborative filtering

Finding “Similar” Users

- Represent users by their rows in the utility matrix

- Two users are similar if their vectors are similar

Similarity Metric

- Jaccard similarity measure

- Cosine similarity measure

- Pearson correlation coefficient

User-user collaborative filtering

Let r_x be the vector of user x 's ratings

$r_x = [1, _, _, 1, 3]$

$r_y = [1, _, 2, 2, _]$

Similarity Metric

Jaccard similarity measure

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Problem: Ignores the value of the rating

r_x, r_y as sets:

$r_x = \{1, 4, 5\}$

$r_y = \{1, 3, 4\}$

User-user collaborative filtering

Let r_x be the vector of user x 's ratings

$$r_x = [1, _, _, 1, 3]$$

$$r_y = [1, _, 2, 2, _]$$

Similarity Metric

Cosine similarity measure

$$\text{sim}(\mathbf{x}, \mathbf{y}) = \cos(\mathbf{r}_x, \mathbf{r}_y) = \frac{\mathbf{r}_x \cdot \mathbf{r}_y}{\|\mathbf{r}_x\| \cdot \|\mathbf{r}_y\|}$$

r_x, r_y as points:

$$r_x = \{1, 0, 0, 1, 3\}$$

$$r_y = \{1, 0, 2, 2, 0\}$$

Problem: This representation leads to unintuitive results.

User-user collaborative filtering

	HP1	HP2	HP3	TW	SW1	SW2	SW3
<i>A</i>	4			5	1		
<i>B</i>	5	5	4				
<i>C</i>				2	4	5	
<i>D</i>		3					3

Intuitively we want: $\text{sim}(A, B) > \text{sim}(A, C)$

$\text{sim}(A, B) = ?$

$\text{sim}(A, C) = ?$

User-user collaborative filtering

	HP1	HP2	HP3	TW	SW1	SW2	SW3
A	4			5	1		
B	5	5	4				
C				2	4	5	
D		3					3

	HP1	HP2	HP3	TW	SW1	SW2	SW3
A	2/3			5/3	-7/3		
B	1/3	1/3	-2/3				
C				-5/3	1/3	4/3	
D		0					0

Mean-centered utility matrix: subtract the means of each row

- 0: means no information
- Negative rating: means viewers with opposite ratings will have vectors in opposite directions.

User-user collaborative filtering

	HP1	HP2	HP3	TW	SW1	SW2	SW3
<i>A</i>	2/3			5/3	-7/3		
<i>B</i>	1/3	1/3	-2/3				
<i>C</i>				-5/3	1/3	4/3	
<i>D</i>		0					0

$\text{sim}(A, B) = ?$

$\text{sim}(A, C) = ?$

User-user collaborative filtering

Finding similar users with overlapping-item mean centering

Mean-centering:

$$r_x = \{1, 0, 0, 1, 3\}$$

$$r_y = \{1, 0, 2, 2, 0\}$$

For each user x, compute mean of each ratings (ignoring missing values)

$$\text{mean}(r_x) = (1+1+3) / 3 = 5/3$$

$$\text{mean}(r_y) = (1+2+2) / 3 = 5/3$$

Subtract this average from each of their ratings (do nothing to the missing values)

$$r_x = \{ 2/3, 0, 0, 2/3, 4/3 \}$$

$$r_y = \{ -2/3, 0, 1/3, 1/3, 0 \}$$

Keep only items they both rate

$$r_x = \{ 2/3, 2/3 \}$$

$$r_y = \{ -2/3, 1/3 \}$$

Then compute cosine similarity

User-user collaborative filtering

Finding similar users with overlapping-item mean centering

Let $\mathbf{r}_x$ be the vector of user x 's ratings

Similarity Metric

Pearson correlation coefficient

S_{xy} = items rated by both users x and y

$$\text{sim}(x, y) = \frac{\sum_{s \in S_{xy}} (r_{xs} - \bar{r}_x)(r_{ys} - \bar{r}_y)}{\sqrt{\sum_{s \in S_{xy}} (r_{xs} - \bar{r}_x)^2} \sqrt{\sum_{s \in S_{xy}} (r_{ys} - \bar{r}_y)^2}}$$

User-user collaborative filtering

Rating predictions

- Let r_x be the vector of user x 's ratings
- Let N be the set of k users most similar to x who have rated item i
- Prediction for item i of user x :

Rate i as the mean of what k -people-like-me rated i

$$r_{xi} = \frac{1}{k} \sum_{y \in N} r_{yi}$$

Rate i as the weighted mean of what k -people-like-me rated i

$$r_{xi} = \frac{\sum_{y \in N} s_{xy} \cdot r_{yi}}{\sum_{y \in N} s_{xy}}$$

Item-Item collaborative filtering

For item i , find other similar items

Estimate rating for item i based on ratings for similar items

Can use same similarity metrics and prediction functions as in user-user model

“Rate i as the mean of my ratings for other items, weighted by their similarity to i ”

$$r_{xi} = \frac{\sum_{j \in N(i;x)} s_{ij} \cdot r_{xj}}{\sum_{j \in N(i;x)} s_{ij}}$$

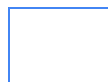
s_{ij} ... similarity of items i and j

r_{xj} ... rating of user x on item j

$N(i;x)$... set items rated by x similar to i

Item-Item CF ($|N|=2$)

		users												
		12	11	10	9	8	7	6	5	4	3	2	1	
movies			4		5			5			3		1	1
		3	1	2			4			4	5			2
			5	3	4		3		2	1		4	2	3
			2			4			5		4	2		4
		5	2					2	4	3	4			5
			4			2			3		3		1	6



- unknown rating



- rating between 1 to 5

Item-Item CF ($|N|=2$)

		Users												
		12	11	10	9	8	7	6	5	4	3	2	1	
movies			4		5			5	?		3		1	1
		3	1	2			4			4	5			2
			5	3	4		3		2	1		4	2	3
			2			4			5		4	2		4
		5	2					2	4	3	4			5
			4			2			3		3		1	6



- estimate rating of movie **1** by user **5**

Item-Item CF ($|N|=2$)

		Users												
		12	11	10	9	8	7	6	5	4	3	2	1	
movies			2/5		7/5			7/5	?		-3/5		-13/5	1
														2
			2	0	1		0		-1	-2		1	-1	<u>3</u>
														4
														5
			7/5			-3/5			2/5		2/5		-8/5	<u>6</u>

Neighbor selection:

Identify movies similar to movie 1, rated by user 5

Here we use Pearson correlation as similarity:

- 1) Subtract mean rating m_i from each movie i
- 2) Compute cosine similarities between rows

Item-Item CF ($|N|=2$)

		users													$\text{sim}(1,m)$
movies		12	11	10	9	8	7	6	5	4	3	2	1		
			4		5			5	?		3		1	1	1.00
		3	1	2			4			4	5			2	
			5	3	4		3		2	1		4	2	<u>3</u>	<u>0.658</u>
			2			4			5		4	2		4	
		5	2					2	4	3	4			5	
			4			2			3		3		1	<u>6</u>	<u>0.768</u>

Compute similarity weights:

$$s_{1,3}=0.658, s_{1,6}=0.768$$

Item-Item CF ($|N|=2$)

		users													
		12	11	10	9	8	7	6	5	4	3	2	1		
movies			4		5			5	2.6		3		1	1	1.00
	3	1	2			4				4	5			2	
		5	3	4		3		2	1			4	2	<u>3</u>	0.658
		2			4			5			4	2		4	
	5	2					2	4	3	4				5	
		4			2			3			3		1	<u>6</u>	0.768

Predict by taking weighted average:

$$r_{1.5} = (0.658 * 2 + 0.768 * 3) / (0.658 + 0.768) = 2.54$$

$$r_{ix} = \frac{\sum_{j \in N(i;x)} s_{ij} \cdot r_{jx}}{\sum s_{ij}}$$

Collaborative filtering

In practice, it has been observed that item-item often works better than user-user

Why?

- Items are simpler, users have multiple tastes

- People are more complex than objects

Collaborative filtering

Pros/Cons of Collaborative Filtering

- + Works for any kind of item
 - No feature selection needed
- Cold Start:
 - Need enough users in the system to find a match
- Sparsity:
 - The user/ratings matrix is sparse
 - Hard to find users that have rated the same items
- First rater:
 - Cannot recommend an item that has not been previously rated
 - New items, Esoteric items
- Popularity bias:
 - Cannot recommend items to someone with unique taste
 - Tends to recommend popular items

Hybrid Methods

- Implement two or more different recommenders and combine predictions
 - Perhaps using a linear model
- Add content-based methods to collaborative filtering
 - Item profiles for new item problem
 - Demographics to deal with new user problem

Remarks & Practical Tips

- Evaluation
- Error metrics
- Complexity / Speed

Evaluation

The diagram shows a 10x6 matrix representing user ratings for movies. The matrix is labeled "users" on the left and "movies" on top. The matrix contains numerical ratings from 1 to 5, with some cells being empty.

1	3	4			
	3	5			5
		4	5		5
		3			
		3			
2			2		2
				5	
	2	1			1
	3			3	
1					

Evaluation

movies

users

1	3	4			
	3	5			5
		4	5		5
		3			
		3			
2			?		?
				?	
	2	1			?
	3			?	
1					

Test Data Set

Evaluating Predictions

Compare predictions with known ratings

- Root-mean-square error (RMSE)

- Precision at top 10: % of relevant items in top 10

Problems with Error Measures

- Narrow focus on accuracy sometimes misses the point
 - Prediction Diversity
 - Prediction Context
 - Order of predictions
- In practice, we care only to predict high ratings:
 - RMSE might penalize a method that does well for high ratings and badly for others

Collaborative Filtering: Complexity

- Too expensive to do at runtime
- We already know how to do this!
 - Near-neighbor search in high dimensions (LSH)
 - Clustering
 - Dimensionality reduction

Tip: Add Data

- Leverage all the data
 - Don't try to reduce data size in an effort to make fancy algorithms work
 - Simple methods on large data do best
- Add more data
 - e.g., add IMDB data on genres